

Kailer Spicer

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Education

Columbia University – GPA: 3.1

New York, NY

B.S. Computer Engineering | Expected May 2027

Relevant Coursework: Circuit Analysis, Advanced Programming in C, Natural Language Processing, Computer Systems

Honors: 2026 Roy G Post Scholar

Washington & Jefferson College – GPA: 3.8

Washington, PA

B.A. Computing and Information Studies | August 2022 – May 2025

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Concepts, Advanced Simulations

Honors: MPSF/6-8 Sports Men's Water Polo All-Academic Team 2023, 2024, CSC Academic All-District At-Large 2023, CWPC All-Academic 2023, Dean's List 2022, 2023, 2024.

Experience

Columbia University — Direct Lab

Research Assistant | New York, NY | January 2026 – Present

- Integrated empirical mobility classifications into computational flow simulations, parameterizing 5 distinct behavioral profiles to increase representation of these groups in new models
- Processed 270 distinct video inputs via the MediaPipe API to extract 33-point skeletal landmarks, yielding over 500 structured training sequences that directly improved the model's classification accuracy.
- Engineered a Graph Convolutional Network (GCN) for automated gait analysis and mobility aid classification, achieving 87.5% validation accuracy using MediaPipe 3D pose landmarks.

Pacific Northwest National Laboratory — Radiological Materials Team

Technical & SULI Intern | Richland, WA | May 2025 – December 2025

- Engineered a lab-ready data pipeline and backend web systems using Python to standardize, manage, and visualize grout datasets, automating reporting and improving research reproducibility.
- Extracted and logged over 1,000 data points from scientific literature into a research database, simultaneously automating data transfers to proprietary modeling software to eliminate manual errors.
- Validated proprietary research software through rigorous QA testing, ensuring data integrity and operational functionality prior to deployment in live research workflows.

Projects

Custom HTTP Web Server (C)

- Designed and implemented a custom HTTP/1.0 web server in C with robust request parsing, standards-compliant status codes, secure URI validation, and fault tolerance to malformed client requests.
- allowed serve to serve static web pages and dynamic content by integrating with a custom database backend, forming a three-tier architecture

Physics-Based Free Throw Simulation (Python, Flask, HTML, CSS)

- Led a 7-member team to develop a physics-based basketball free-throw simulation modeling projectile motion with drag and Magnus forces based on academic literature.
- Integrated the Python simulation with an interactive Flask web interface to enable real-time visualization and user-controlled parameter exploration.

Languages & Skills

Programming Languages: Python, Java, C, C++, SQL, HTML, CSS

Frameworks & Libraries: Pandas, Flask, JavaFX, Matplotlib, Pytorch, MediaPipe

Databases: NoSQL, Sqlite, Postgres, MongoDB

Tools: Git, Tableau, VBA, Linux, CLIs, Vim, Colab

Systems & Concepts: Data Pipelines, Backend Development, QA Testing, HPC, Data Modeling, ML, NLP, Simulations, LLM Training, Video Extraction